Phase – II CAPSTONE PROJECT

EduPredictAI

AI-Powered Student Performance & At-Risk Early Warning System

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Github: https://github.com/vanshikasharma48/Capstone-Project-New-Age-programming

1. Executive Summary & Platform Overview

The Dashboard Overview serves as the administrative command center for EduPredictAI, greeting users upon login and displaying core institutional metrics such as total active students (5), overall average score (71%), average attendance (82%), and high-risk student alerts. It includes a monthly academic progress tracker and a student risk overview showing that 40% of the cohort is at high risk (comprising 2 high-risk, 1 medium-risk, and 2 low-risk students). Additionally, it features a recent student snapshot preview table displaying individual records with student IDs, classes, attendance percentages, academic scores, and risk classifications for quick administrative monitoring.

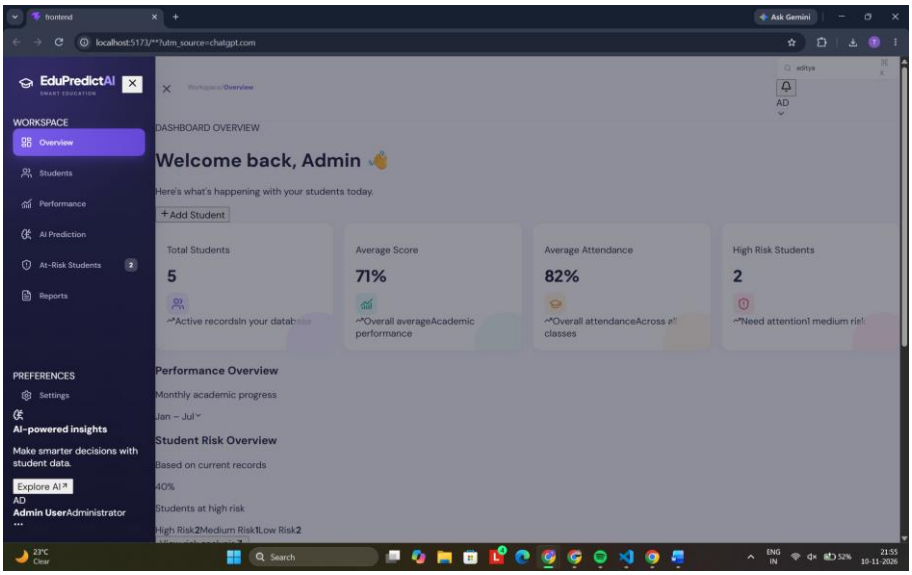


Fig 1.

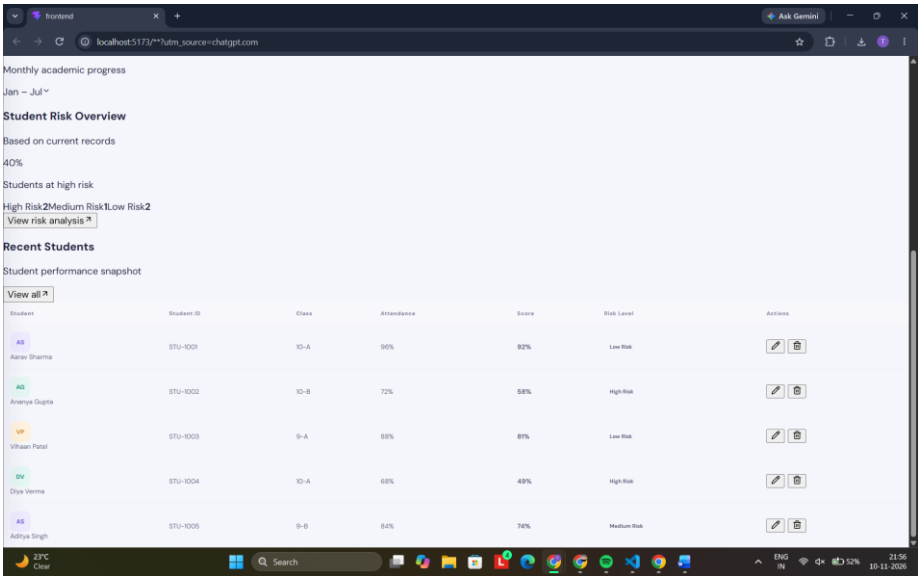


Fig 2.

Student Management & Directory

The **Student Management & Directory** section functions as the core database interface of EduPredictAI, enabling administrators to seamlessly oversee student records, academic scores, and attendance metrics. At the top of the directory, summary cards categorize the entire student cohort by risk levels—tracking **Total Students (5)**, **Low Risk (2)**, **Medium Risk (1)**, and **High Risk (2)**. The interface provides intuitive management tools including an **"+ Add Student"** button for onboarding new records, risk-level filters, and an **"Export CSV"** function for data portability. The comprehensive student table lists all active records complete with student names, IDs, classes, attendance rates, scores, risk badges, and direct edit/delete action controls

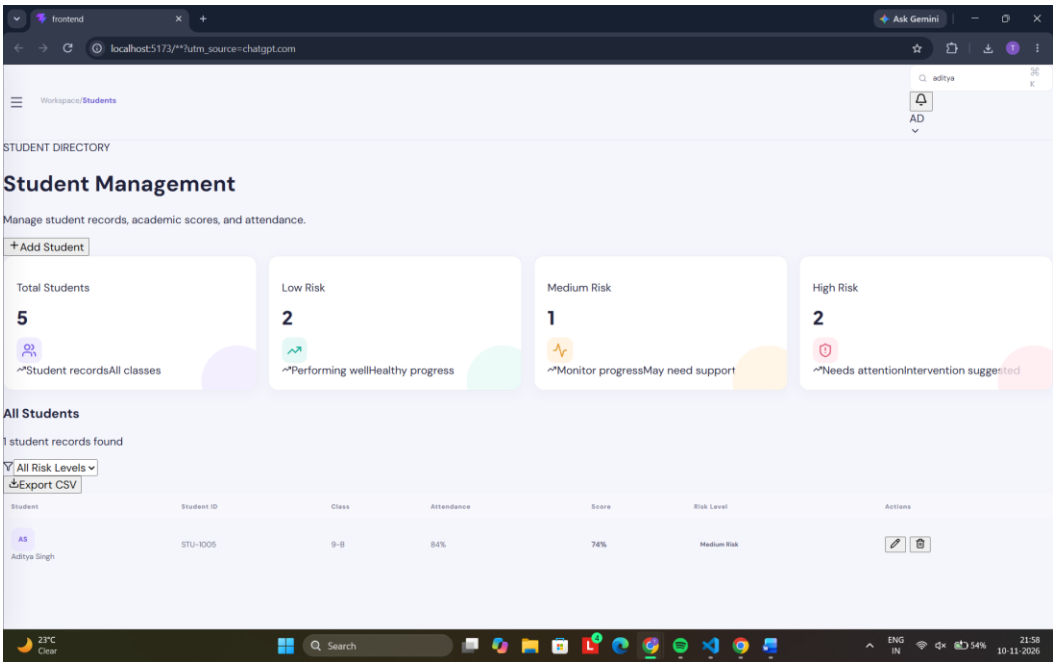


Fig 3.

Performance Analytics & Score Distribution

The Performance Analytics section provides a detailed breakdown of academic results across subjects, illustrating student performance tiers and rankings. It categorizes student achievement into four distinct score ranges—Excellent (80–100%) with 2 students, Good (70–79%) with 1 student, Average (60–69%) with 0 students, and Needs Improvement (Below 60%) with 2 students. Additionally, it features a Top Performing Students leaderboard ranked by academic score, highlighting leaders such as Aarav Sharma (92%), Vihaan Patel (81%), and Aditya Singh (74%), alongside students requiring academic focus like Ananya Gupta (58%) and Diya Verma (49%). The module also ensures data transparency by noting that metrics like average scores and leaderboards derive from saved student records.

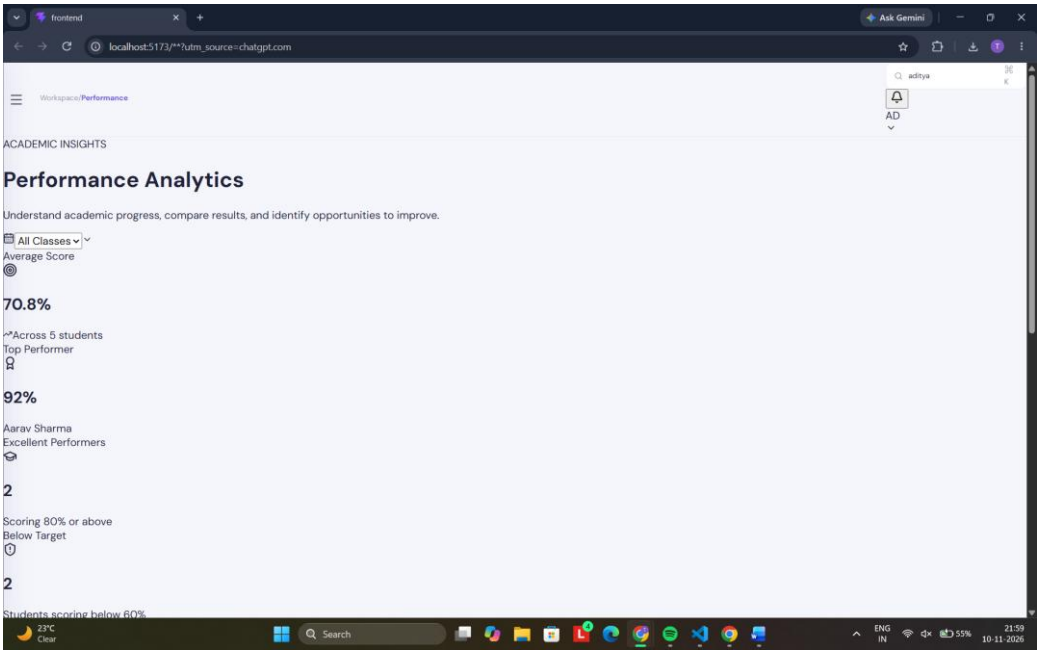


Fig 4.

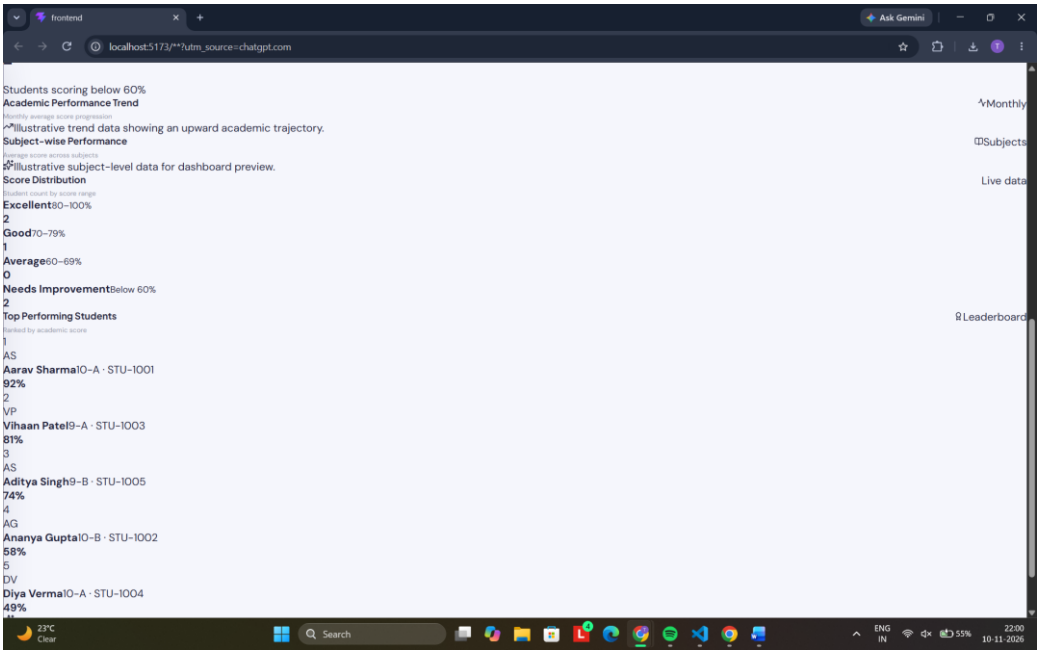


Fig 5.

Reports & Insights (Academic Reporting)

The Reports & Insights section serves as the centralized academic reporting hub of EduPredictAI, designed to generate comprehensive academic summaries, student performance reports, and class-level insights. The interface provides a streamlined administrative layout featuring an "Export Student Data" action button, allowing educators to instantly download and export performance metrics for external evaluation and documentation.

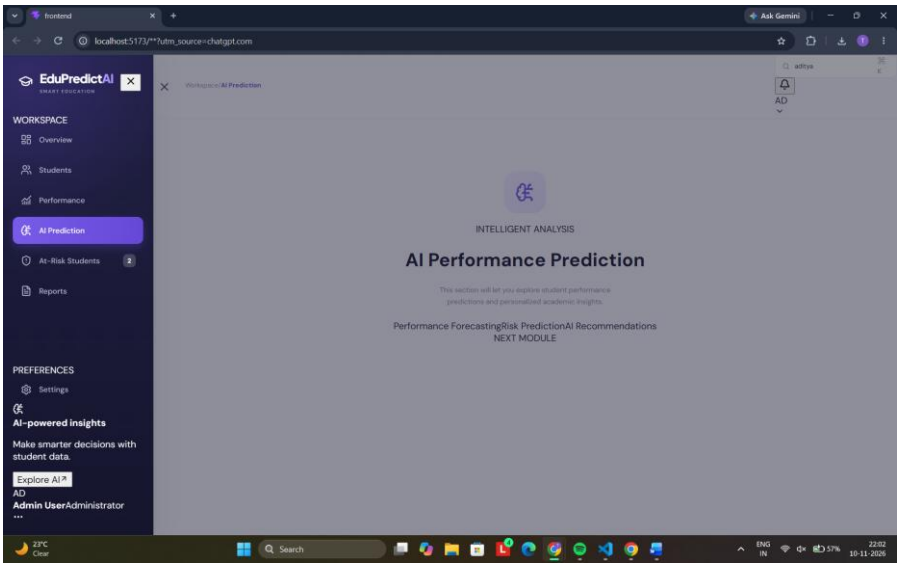
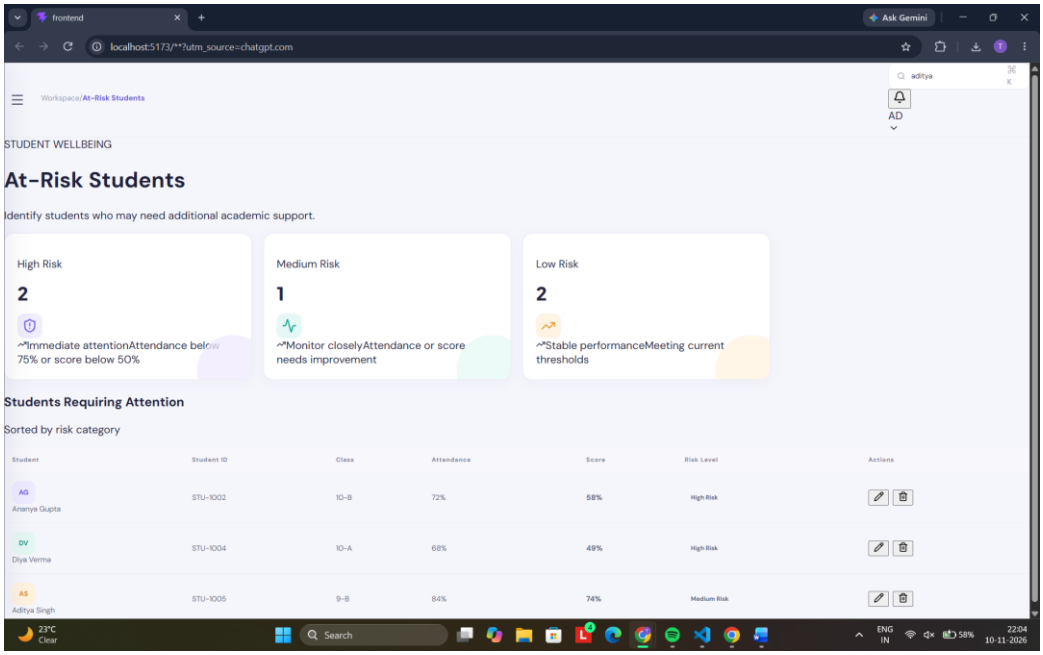


Fig 7.

Academic Insights & Performance Analytics

The **Academic Insights** section within Performance Analytics offers a high-level statistical evaluation to track academic progress and identify areas needing improvement across all classes. The module displays an overall **Average Score of 70.8%** calculated across the 5 enrolled students. It highlights top-tier performance metrics, recognizing **Aarav Sharma** as the **Top Performer** with a highest score of **92%**. Furthermore, the system categorizes student outcomes into clear performance brackets: **2 Excellent Performers** scoring 80% or above, and **2 Below Target** students scoring below 60%, enabling educators to quickly benchmark class efficiency and pinpoint learning gaps.



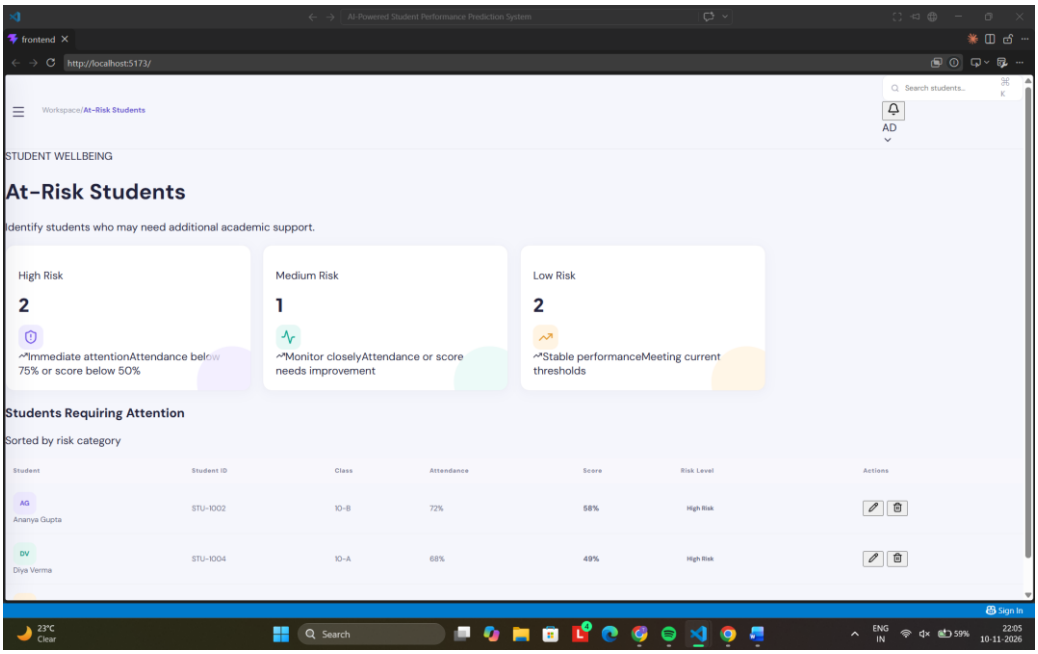
Dynamic Student Search & Filter Operations

The **Student Management Directory** incorporates real-time search and filtering capabilities to manage individual student profiles efficiently. As demonstrated during a targeted search query (e.g., searching for *"aditya"*), the system dynamically filters the dataset to instantly locate specific student records. The filtered result highlights **Aditya Singh** (STU-1005 | Class 9-B), displaying an attendance rate of **84%**, an academic score of **74%**, and an automatically assigned **Medium Risk** classification badge. Despite active search filters, the top analytical summary cards continuously reflect full institutional health metrics—maintaining total student counts (5) alongside Low (2), Medium (1), and High Risk (2) distributions for complete context.

Student Wellbeing & At-Risk Identification

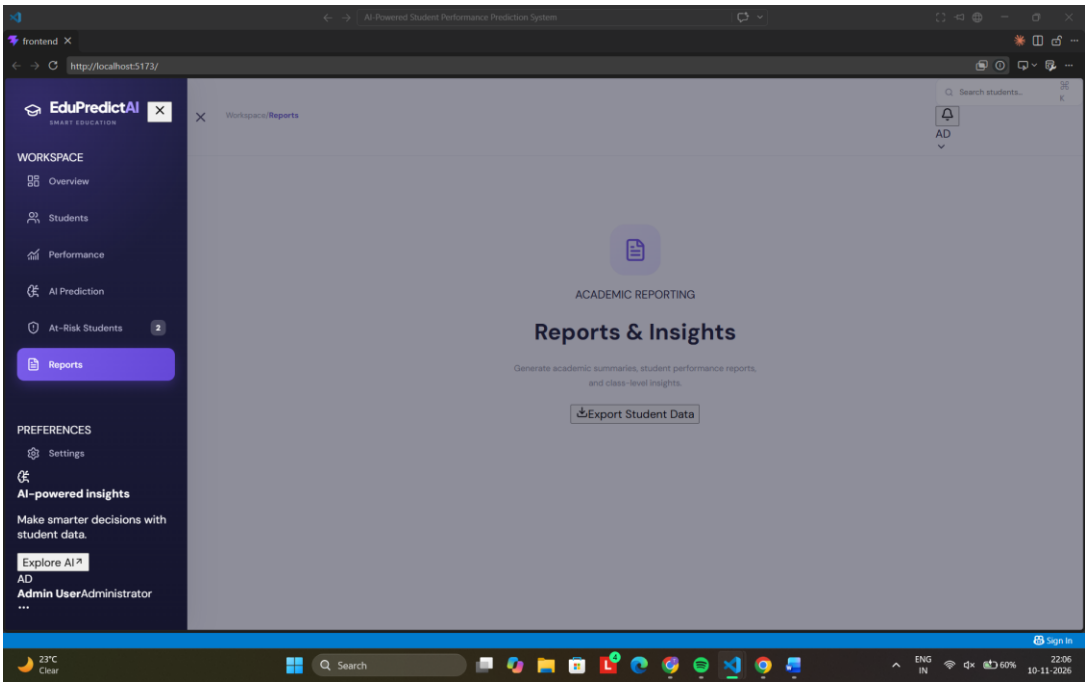
The Student Wellbeing (At-Risk Students) section focuses on identifying students requiring immediate or supplementary academic support based on standardized risk criteria. The interface categorizes students into three distinct thresholds: High Risk (2 students) for those with attendance below 75% or scores below 50%, Medium Risk (1 student) requiring close monitoring due to borderline scores or attendance, and Low Risk (2 students) meeting all academic baseline targets. The *Students Requiring Attention* table automatically sorts and prioritizes cases needing intervention, explicitly flagging students such as Ananya Gupta (72% attendance, 58% score) and Diya Verma (68% attendance, 49% score) under High Risk, and Aditya Singh (84% attendance, 74%

score) under Medium Risk.



AI Performance Prediction & Recommendations

The **AI Performance Prediction** module functions as the intelligent analytical core of EduPredictAI, designed to deliver forward-looking performance forecasting, automated risk prediction, and personalized AI recommendations. By evaluating individual student historical trends, score distributions, and attendance behavior, this module empowers educators to move from reactive grading to proactive academic guidance, allowing timely interventions before major institutional examinations.



Conclusion and Future Scope

1. Conclusion

The EduPredictAI – AI-Powered Student Performance Prediction System demonstrates a complete, modern, and SaaS-inspired educational technology platform designed to streamline student data management and academic tracking. The system features a responsive and intuitive user interface built with React, Vite, and Tailwind-based components, seamlessly connected to backend operations and live database tracking. Key modules—including the Dashboard Overview, Student Management Directory, Performance Analytics, Student Wellbeing (At-Risk Tracking), and Reports & Insights—effectively empower educators to monitor live student metrics, track attendance and scores, and identify at-risk learners early through automated risk categorization. The platform successfully bridges the gap between raw educational data and actionable administrative interventions, shifting academic management from a reactive approach to a proactive model.

2. Future Scope

While the platform currently provides full system navigation, real-time data filtering, and active database integration, several advanced enhancements are planned for future updates:

- Advanced Machine Learning Pipelines: Integrating higher-order predictive ML models (e.g., deep learning models and complex regression algorithms) to enhance long-term score forecasting accuracy and automate personalized student growth recommendations.
- Interactive Data Visualizations: Expanding the analytics dashboard with advanced, interactive visual charting tools (like dynamic heatmaps and multi-subject comparison radar charts) for deeper academic insights.
- Cross-Platform Native Mobile Application: Finalizing and releasing the dedicated Android native application (built with Kotlin and Jetpack Compose) to allow teachers and parents to receive instant push notifications for high-risk student alerts on mobile devices.
- Automated Parent Communication System: Adding an integrated SMS and email notification system to automatically dispatch weekly attendance and performance summaries directly to guardians.